

Thomas Lilygren

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EDUCATION

Harvey Mudd College, Claremont, CA

Graduated May 2026

B.S. Engineering with EE focus - Graduated with Distinction (GPA: 3.55/4.00), Dean's List (all semesters)

- **Activities:** Experimental Engineering Proctor, Engineering Mathematics Grader and Tutor, Calculus Tutor
- **Relevant Coursework:** Microprocessor Design and Application | Analog Circuit Design | Digital Signal Processing | Advanced Systems Engineering | Digital Electronics and Computer Engineering | Experimental Engineering | Continuum Mechanics | Engineering Mathematics
- **Projects:** Constructed an autonomous underwater vehicle with ballasting and proportional depth control systems | Implemented a RISC-V Multicycle Processor in SystemVerilog | Barcode Scanner Audio Synthesizer | Stable High-Gain BJT Op-Amp Design and Validation | Fast Music Retrieval Under Time Warping | Developed a game of "Snake" with AI capability using Java | Programmed a game of "Mastermind" with AI functionality using Python

IES Abroad, Barcelona, Spain

August 2024 - December 2024

Full-time internship program

RESEARCH PROJECTS

FAA/Aerospace Corporation Clinic, Harvey Mudd College

August 2025 - May 2026

Team Lead

- Led a team of 6 multi-disciplinary engineers in designing payloads for the next generation of GPS.
- Directed an in-depth GNSS trade space literature review to generate state-of-the-art GPS overlay proposals.
- Coordinated communication and planning with professional aerospace engineers.
- Advised FAA decision-making in a Concurrent Design Center Study hosted by the Aerospace Corporation

UC Berkeley Clinic, Harvey Mudd College

January 2025 - May 2025

Team Member

- Upgraded communication and localization protocols for a fleet of 100+ low-cost Aerosol Black Carbon Detectors
- Redesigned Arduino code to substitute 2G data transmission with Wi-Fi and comply with the Arduino Nano ESP32
- Implemented CSRF authentication and automated sensor localization

Advanced Science Investigations, Los Altos, CA

July 2020 - June 2021

Student Researcher

- Developed a novel hafnium-based metal-organic framework for trivalent arsenic adsorption in fresh water.
- Analyzed scientific publications, consulted with professional scientists, structured experiments, practiced scanning electron microscopy, and presented research in county and state science fairs.

WORK EXPERIENCE

Stanford Graduate School of Business, Palo Alto, CA

May 2025 - August 2025 | May 2026 - August 2026

Stanford Executive Program Assistant

- Supported the Stanford Executive Program through classroom setup, event coordination, and program communications, working closely with faculty and full-time program staff.
- Performed administrative tasks such as assembling classroom materials, entering data, and preparing documents to ensure smooth and efficient program delivery for 200+ C-suite executives.

THEKER Robotics, Barcelona, Spain

August 2024 - December 2024

Electrical Engineering Intern

- Designed custom PCBs with Autodesk Fusion to streamline THEKER's electric box layout.
- Developed wiring schematics to expedite manufacturing process, document the iterative process, and aid new hires.
- Constructed several electric boxes for commercial implementation.

AWARDS

First Place – 2021 Synopsys Silicon Valley Science and Technology Championship | **First Place** – American Chemical Society, Silicon Valley | **Qualifier** – 2021 California Science and Engineering Fair | **Third Place** – Society for the Advancement of Material and Process Engineering | **Third Place** – ASM International - A Society for Materials, Santa Clara

SKILLS

- **Software:** SystemVerilog, C/C++, Python, Java, Arduino IDE, LT Spice, MATLAB, Simulink, Autodesk Fusion, SolidWorks, Jupyter Notebook, Modbus, Questa, Quartus, Lattice Radiant, Segger, Github, Excel
- **Hardware:** FPGAs, Microcontrollers, Datasheets, Oscilloscope, Soldering, 3D Printing, PCB Design
- **Languages:** Spanish (Proficient)

INTERESTS

LEGOs, Poker, Clarinet, Mountain Biking, Travel, Golf, Climbing, Ping Pong, Drawing, Boxing, Reading, Philosophy